Software Requirements Specification for **Full-stack based lung cancer risk prediction using ML,DL and QML**

Prepared by:
Shashank G
Nithish N
Pruthvi R
Shrinidhi R

RV University
Date: 13.03.2026

CONTENTS

I Introduction 2
- I-A Purpose and scope of srs . . 2
- I-B Document Conventions . . . 2
- I-C Intended Audience and Reading Suggestions 2
- I-D Product Scope 2
- I-E References 3

II Overall Description 3
- II-A Product Perspective 3
- II-B Product Functions 3
- II-C User Classes and Characteristics 3
- II-D Operating Environment . . . 4
- II-E Design and Implementation Constraints 4
- II-F User Documentation 4
- II-G Assumptions and Dependencies 5

III External Interface Requirements 5
- III-A User Interfaces 5
- III-B Hardware Interfaces 5
- III-C Software Interfaces 5
- III-D Communications Interfaces . 6

IV System Features 6
- IV-A System Feature 1: User Login 6
  - IV-A1 Description and Priority 6
  - IV-A2 Stimulus/Response Sequences 6
  - IV-A3 Functional Requirements . . 6

V Other Nonfunctional Requirements 6
- V-A Performance Requirements . 6
- V-B Safety Requirements 6
- V-C Security Requirements 7
- V-D Software Quality Attributes . 7
- V-E Business Rules 7

VI Other Requirements 7

Appendix A: Glossary 7

Appendix B: Analysis Models 7
- B-A Data Flow Diagram (DFD) . 7
- B-B UML Use Case Diagram . . 7
- B-C Entity Relationship (ER) Diagram 7

Appendix C: To Be Determined 7

I. Introduction

A. Purpose and scope of srs

The purpose of the software is to develop a Lung Cancer Risk Prediction System that helps predict the likelihood of lung cancer in individuals based on various health-related parameters such as smoking habits, coughing, wheezing, chest pain, and other symptoms. The system uses machine learning and deep learning techniques to analyze the input data and provide a prediction of whether a person is at risk of lung cancer. The software also provides a simple web-based interface where users can enter their health information and receive an instant prediction result. This helps in early risk assessment and awareness, allowing users to seek medical advice at an early stage. This SRS document describes the functional and non-functional requirements of the Lung Cancer Prediction System. It outlines the system's overall structure, including the user interface, data processing, machine learning model integration, and prediction output. The document defines how the system collects user input, processes the data using trained models, and displays prediction results through a web application. It also specifies the software environment, system constraints, and performance expectations required for the successful operation of the system. The SRS serves as a guideline for developers, testers, and stakeholders to understand the system's functionality and ensure that the software meets the intended objectives.

B. Document Conventions

This Software Requirements Specification (SRS) follows standard documentation conventions to ensure clarity and consistency throughout the document. Headings and subheadings are used to organize the content into sections and subsections. Bold text is used to highlight important terms, section titles, and key concepts. Bullet points and numbered lists are used to present requirements and explanations in a structured and easy-to-read format. Requirement statements are written in clear and precise language to avoid ambiguity. Higher-level requirements may define the general functionality of the system, while detailed requirements provide specific descriptions of how the system should behave.

Abbreviations and Acronyms

The document may use several abbreviations and acronyms for simplicity and readability. Some commonly used ones include:

SRS Software Requirements Specification
ML Machine Learning
DL Deep Learning
UI User Interface
CSV Comma Separated Values
API Application Programming Interface

These abbreviations are used consistently throughout the document to make the specification concise and easier to understand.

C. Intended Audience and Reading Suggestions

This Software Requirements Specification (SRS) is intended for several types of readers involved in the development and use of the Lung Cancer Prediction System. Developers use this document to understand the system architecture, functional requirements, and implementation details required to build the software. Testers refer to the document to verify that the system meets all specified requirements and to design appropriate test cases. Project managers use the SRS to monitor project progress, manage resources, and ensure that the development follows the planned objectives. End users such as healthcare professionals or individuals using the system can refer to the document to understand the system's functionality and how predictions are generated. The document is organized into multiple sections. The introduction section provides an overview of the system and its objectives. The overall description section explains the system perspective, features, and constraints. The specific requirements section describes the detailed functional and non-functional requirements of the system. Readers are recommended to begin with the introduction and overall description sections to understand the system context, and then proceed to the specific requirements sections that are most relevant to their role.

D. Product Scope

The Lung Cancer Risk Prediction System is a software application designed to predict the likelihood of lung cancer based on various

health-related parameters such as smoking habits, coughing, chest pain, wheezing, and other symptoms. The system uses machine learning and deep learning models to analyze the input data provided by users and generate a prediction indicating whether the individual may be at risk of lung cancer.

The primary goal of this software is to assist in early risk assessment and increase awareness about potential lung cancer symptoms. By providing quick and data-driven predictions through a user-friendly web interface, the system can help individuals and healthcare professionals make informed decisions and seek timely medical consultation. The software also demonstrates the practical application of artificial intelligence techniques in healthcare, supporting the broader objective of using technology for improved health monitoring and decision support.

E. References

1 IEEE Computer Society, IEEE Recommended Practice for Software Requirements Specifications (IEEE Std 830-1998), IEEE, 1998.

2 S. Anand et al., Lung Cancer Risk Prediction System – Project Design Document, Version 1.0, 2026.

3 F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.

4 M. Abadi et al., "TensorFlow: Large-Scale Machine Learning on Heterogeneous Systems," Google Research, 2015.

5 A. Ronacher, Flask Web Development Documentation, Pallets Project, 2010.

II. Overall Description

A. Product Perspective

The Lung Cancer Risk Prediction System is a standalone web-based application designed to analyze user health data and predict the likelihood of lung cancer using machine learning and deep learning techniques. The system is developed as a self-contained product that integrates data preprocessing, model prediction, and a user-friendly interface into a single platform.

The application consists of three main components: a web interface, a backend server, and a machine learning prediction model. The web interface allows users to enter health-related parameters such as smoking habits, coughing, wheezing, and chest pain. The backend server processes the input data, performs necessary preprocessing, and sends the data to the trained deep learning model. The prediction model then analyzes the data and returns the result indicating the possible risk of lung cancer, which is displayed to the user through the web interface. This system does not depend on any larger external system and operates independently. However, it can be integrated with healthcare decision support systems or hospital management systems in the future to assist healthcare professionals in early risk assessment and patient monitoring. The system interacts with external resources such as the lung cancer dataset, machine learning libraries, and model files required for prediction..

B. Product Functions

The Lung Cancer Risk Prediction System performs several key functions to provide accurate and efficient risk assessment. The major system functions include:

- **User Input Collection:** Allows users to enter health-related information such as smoking habits, coughing, wheezing, chest pain, and other symptoms through the web interface.
- **Data Processing:** Processes the input data by applying preprocessing techniques such as encoding and feature scaling before sending it to the prediction model.
- **Prediction and Result Generation:** Uses the trained machine learning or deep learning model to analyze the processed data and generate a prediction indicating the likelihood of lung cancer.
- **Result Display:** Displays the prediction outcome to the user through a clear and simple web interface.
- **Report Generation:** Generates a summary of the prediction results that can be viewed or stored for future reference.

C. User Classes and Characteristics

The Lung Cancer Risk Prediction System is designed to be used by different types of users.

Each user class has different access levels and responsibilities within the system.

- **Admin:** The admin has the highest level of access in the system. The admin is responsible for managing the system, monitoring system usage, maintaining the dataset or model configurations, and overseeing overall system functionality. Admin users are expected to have moderate to high technical knowledge.
- **Registered User:** Registered users are individuals who create an account in the system to access prediction features. They can input health-related data, obtain lung cancer risk predictions, and view their results. These users require basic computer knowledge to interact with the web interface.
- **Guest User:** Guest users can access limited features of the system without creating an account. They may view general information about lung cancer prediction and the purpose of the system but may have restricted access to prediction functionalities.

D. Operating Environment

The Lung Cancer Risk Prediction System operates in a web-based environment and can run on multiple platforms. The software environment required for the system is described below:

- **Operating System:** The system can run on Windows, Linux, or macOS operating systems.
- **Backend:** The backend of the application is implemented using Python (Flask) or Node.js to handle data processing, model execution, and server-side operations.
- **Frontend:** The user interface is developed using HTML, CSS, and JavaScript to provide an interactive web-based platform for users to input health data and view prediction results.
- **Database:** The system can use databases such as MySQL, MongoDB, or SQLite for storing user information, prediction results, and system data.

E. Design and Implementation Constraints

The development and implementation of the Lung Cancer Risk Prediction System are subject to certain constraints that limit the available design and development options. These constraints are described below:

- **Programming Languages and Frameworks:** The system is developed using Python for machine learning model implementation and backend services, while HTML, CSS, and JavaScript are used for the frontend interface.
- **Machine Learning Libraries:** The system relies on machine learning and deep learning libraries such as Scikit-learn, TensorFlow, and Keras for model training and prediction.
- **Hardware Limitations:** The system is designed to run on standard personal computers without requiring specialized hardware such as GPUs.
- **Database Technologies:** The application uses lightweight databases such as SQLite for storing data, which may limit scalability for large-scale deployments.
- **Security Considerations:** User data must be handled securely, and proper validation and access control mechanisms must be implemented to protect sensitive information.
- **Web Application Constraints:** The system must be compatible with modern web browsers and maintain responsive performance for users accessing the application online.

F. User Documentation

The Lung Cancer Risk Prediction System will include several user documentation components to help users understand and effectively use the software.

- **User Manual:** Provides detailed instructions on how to use the system, including entering health data, running predictions, and interpreting the results.
- **Installation Guide:** Describes the steps required to install and configure the software, including system requirements, dependency installation, and setup procedures.
- **Online Help:** Offers quick guidance and explanations within the web interface to assist users while interacting with the system.

G. Assumptions and Dependencies

The following assumptions and dependencies are considered during the development and operation of the Lung Cancer Risk Prediction System:

- **Internet Connection:** The system assumes that users have a stable internet connection to access the web-based application and perform prediction tasks.
- **Web Browser Availability:** Users must have access to a modern web browser such as Google Chrome, Mozilla Firefox, or Microsoft Edge to interact with the system interface.
- **Software Dependencies:** The system depends on external libraries and frameworks such as Python, Flask, TensorFlow, and Scikit-learn for machine learning model execution.
- **Dataset Availability:** The prediction model assumes that the lung cancer dataset used for training and testing is accurate and available during development.

III. External Interface Requirements

A. User Interfaces

The Lung Cancer Risk Prediction System provides a user-friendly web interface that allows users to enter health-related information and view prediction results. The interface includes an input form, result display section, and validation messages to guide the user during interaction.

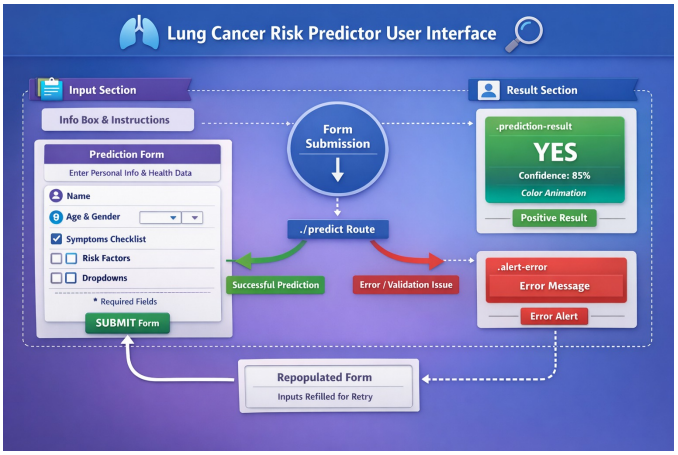

Fig. 1. User Interface Workflow of the Lung Cancer Risk Prediction System

B. Hardware Interfaces

The Lung Cancer Risk Prediction System interacts with standard computer hardware components required to operate the web application and perform predictions.

- **Input Devices:** The system supports standard input devices such as a keyboard and mouse that allow users to enter health-related information into the web interface.
- **Display Devices:** The prediction results and user interface are displayed through standard monitor or screen devices.
- **Processing Unit:** The application runs on a standard CPU capable of executing Python-based machine learning models and handling backend processing.
- **Storage Devices:** Local or cloud-based storage devices are used to store datasets, trained model files, and system configuration data.
- **Network Interface:** A network interface such as Wi-Fi or Ethernet is required to allow users to access the web-based application through the internet.

C. Software Interfaces

The Lung Cancer Risk Prediction System interacts with several software components required for data processing, model execution, and system functionality.

- **Database Interface:** The system connects to a database such as SQLite, MySQL, or MongoDB to store user data, prediction results, and system records.
- **Machine Learning Modules:** The application uses machine learning libraries such as Scikit-learn and TensorFlow/Keras to perform data preprocessing, model training, and prediction tasks.
- **Backend Framework:** The system uses a backend framework such as Flask (Python) to handle user requests, process data, and communicate with the prediction model.
- **Frontend Technologies:** The user interface is built using HTML, CSS, and JavaScript, which communicate with the backend server to send input data and receive prediction results.

- **External APIs:** If required, the system can integrate with external healthcare APIs or data services to enhance prediction accuracy or provide additional medical information.

D. Communications Interfaces

The Lung Cancer Risk Prediction System communicates through standard web-based communication protocols to enable interaction between the user interface, backend server, and machine learning modules.

- **HTTP/HTTPS Protocol:** The system uses HTTP or secure HTTPS protocols for communication between the client (web browser) and the backend server.
- **REST APIs:** RESTful APIs are used to handle requests and responses between the frontend interface and the backend application for data submission and prediction results.
- **Web Browser Communication:** Users interact with the system through web browsers such as Google Chrome, Mozilla Firefox, or Microsoft Edge.
- **Data Format:** Data exchanged between the client and server is typically formatted in JSON or form-data format for efficient processing.
- **Security and Encryption:** Secure communication protocols such as HTTPS are used to protect user data during transmission.

IV. System Features

A. System Feature 1: User Login

1) Description and Priority: **Description:** This feature allows users to securely log into the Lung Cancer Risk Prediction System using their credentials. After successful authentication, users can access system functionalities such as entering health information and viewing prediction results. **Priority:** High

2) Stimulus/Response Sequences:

- **Stimulus:** The user enters login credentials (username and password) in the login form.
- **Response:** The system receives the credentials and sends them to the authentication module for validation.
- **Stimulus:** The system checks the entered credentials against the stored user data in the database.
- **Response:** If the credentials are valid, the system grants access to the user dashboard.
- **Response:** If the credentials are invalid, the system denies access and displays an appropriate error message.

3) Functional Requirements:

- **REQ-1:** The system shall allow users to log into the application using valid login credentials.
- **REQ-2:** The system shall validate the entered username and password with the stored user data in the database.
- **REQ-3:** The system shall display an appropriate error message if the login credentials are invalid.
- **REQ-4:** The system shall grant access to authorized users after successful authentication.
- **REQ-5:** The system shall prevent unauthorized users from accessing protected features.

V. Other Nonfunctional Requirements

A. Performance Requirements

The Lung Cancer Risk Prediction System should provide efficient performance to ensure a smooth user experience and timely prediction results.

- **System Response Time:** The system should generate prediction results and respond to user requests within 2 seconds under normal operating conditions.
- **Concurrent Users:** The system should support up to 100 users simultaneously without significant degradation in performance.
- **Processing Efficiency:** The prediction model should process user input data quickly to provide real-time or near real-time results.

B. Safety Requirements

The Lung Cancer Risk Prediction System should include safety measures to prevent data loss and ensure reliable system operation.

- **Data Protection:** The system shall implement mechanisms to prevent accidental loss or corruption of user data during processing and storage.

- **System Recovery:** The system shall provide recovery mechanisms such as data backups or restart procedures in case of system failure.
- **Error Handling:** The system shall handle unexpected errors gracefully and display appropriate messages without affecting system stability.

C. Security Requirements

The Lung Cancer Risk Prediction System must ensure that user data and system resources are protected from unauthorized access.

- **User Authentication:** The system shall require users to authenticate themselves using valid login credentials before accessing protected features.
- **Secure Password Storage:** User passwords shall be stored securely using hashing or encryption techniques to prevent unauthorized access.
- **Access Control:** The system shall restrict access to sensitive system functions based on user roles and permissions.
- **Data Protection:** The system shall protect user data during transmission using secure communication protocols such as HTTPS.

D. Software Quality Attributes

The Lung Cancer Risk Prediction System should satisfy several quality attributes to ensure reliability, efficiency, and ease of use.

- **Reliability:** The system should operate consistently and provide accurate prediction results with minimal system failures.
- **Maintainability:** The system should be designed with modular code and clear documentation so that updates, bug fixes, and improvements can be easily implemented.
- **Scalability:** The system should be capable of handling an increasing number of users or larger datasets without significant degradation in performance.
- **Usability:** The system should provide a simple and intuitive user interface that allows users to easily enter data and understand prediction results.

E. Business Rules

The Lung Cancer Risk Prediction System follows certain operational policies and permission rules to regulate system usage and ensure proper access control.

- **User Access Control:** Only authenticated users are allowed to access the system's prediction features.
- **Role-Based Permissions:** Different user roles such as Admin and Registered User have different access privileges within the system.
- **Data Usage Policy:** User health information entered into the system should only be used for prediction purposes and must not be shared with unauthorized parties.
- **System Usage Policy:** Users must follow the system guidelines and provide accurate health information to obtain reliable prediction results.

VI. Other Requirements

Include database requirements, legal constraints, or regulatory requirements.

Appendix A
Glossary

- API – Application Programming Interface
- AI – Artificial Intelligence

Appendix B
Analysis Models

The following analysis models illustrate the system structure, data flow, and relationships within the Lung Cancer Risk Prediction System.

- Data Flow Diagram
- UML Diagram
- Entity Relationship Diagram

A. Data Flow Diagram (DFD)

B. UML Use Case Diagram

C. Entity Relationship (ER) Diagram

Appendix C
To Be Determined

- TBD-1
- TBD-2

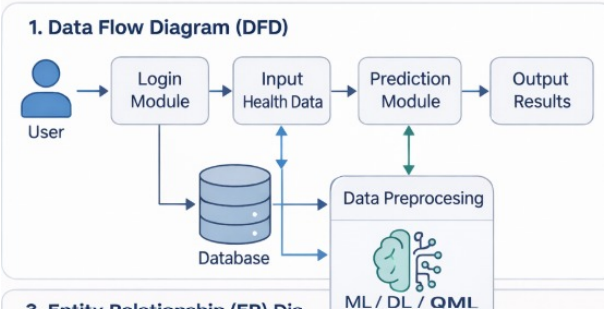

Fig. 2. Data Flow Diagram of Lung Cancer Risk Prediction System using ML, DL and QML Models

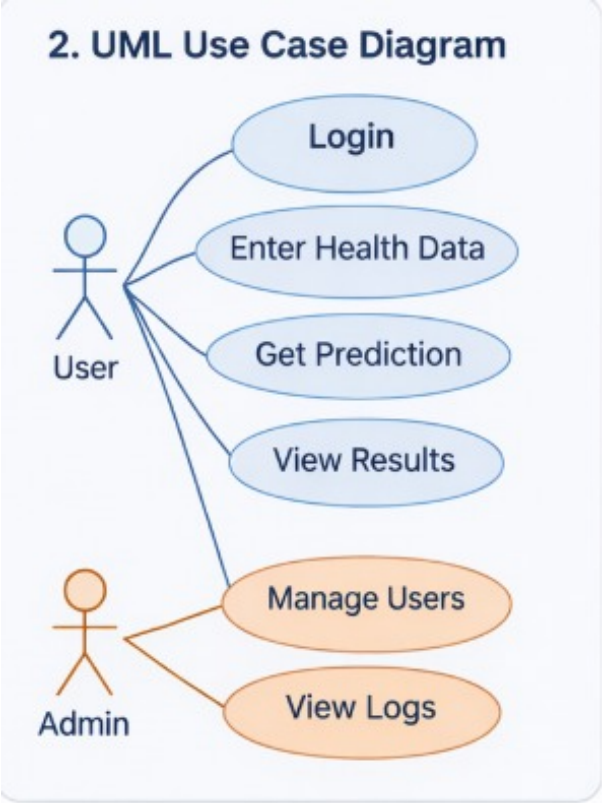

Fig. 3. UML Use Case Diagram of Lung Cancer Risk Prediction System

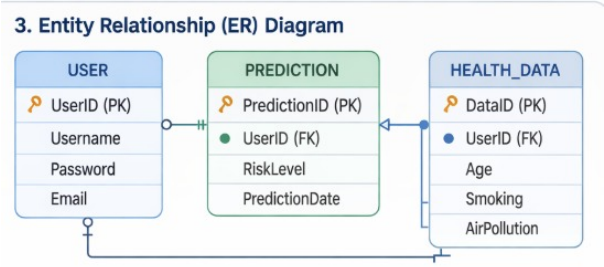

Fig. 4. Entity Relationship Diagram of Lung Cancer Risk Prediction System